# 2_10xMouse_RA_STARsolo

August 31, 2026

[1]:
```
library(Seurat)
library(ggplot2)
library(pheatmap)
library(dplyr)
library(RColorBrewer)
getwd()
dir.create("figures_10xMouse_RA")
dir.create("data_10xMouse_RA")
colorDict = c("2CLC"="#88535A",
              "Zscan4+"="#EF8264",
              "Pluri"="#F2CC8F")
```

```
Loading required package: SeuratObject

Loading required package: sp


Attaching package: 'SeuratObject'


The following objects are masked from 'package:base':

    intersect, t



Attaching package: 'dplyr'


The following objects are masked from 'package:stats':

    filter, lag


The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

'/mnt/TEdata/TEbenchmarking/jupyterNotebooks/realDatasets'

```
Warning message in dir.create("figures_10xMouse_RA"):
"'figures_10xMouse_RA' already exists"
Warning message in dir.create("data_10xMouse_RA"):
"'data_10xMouse_RA' already exists"
```

[2]:
```
dataset_id <- "10xMouse_RA"
sample <- "LIF"
path <- paste0("/mnt/volume_1p5T/results/STARsolo_EM_TE/",dataset_id,"/
↪",sample,"/TE_Solo.out/Gene/")

# get filtered barcodes
STAR_path <- paste0("/mnt/TEresults/snakemake_results/results/STARoutdir/
↪",dataset_id,"/",sample,"/best_Solo.out/Gene")
filteredBarcodes <- read.table(paste0(STAR_path, "/filtered/barcodes.tsv"))$V1␣
↪# read barcodes seleced by STARsolo

STARsolo_TE_EM_mat <- Seurat::ReadMtx(
        mtx = paste0(path, "/raw/UniqueAndMult-EM.mtx"),
        cells = paste0(path, "/raw/barcodes.tsv"),
        features = paste0(path, "/raw/features.tsv")
    ) # read matrix
STARsolo_TE_EM_mat <- STARsolo_TE_EM_mat[, filteredBarcodes] # keep only␣
↪barcodes passing thresholds

nCells <- length(filteredBarcodes)
thrMinCells <- round(nCells * 0.02)
thrMinCells

objTE_STARsolo <- Seurat::CreateSeuratObject(STARsolo_TE_EM_mat,
        project = "STARsolo_TE_EM",
        min.cells = thrMinCells, min.features = 0
    )


objTE_STARsolo
```

[ ]:
```
annotation_stellarscope <- read.table("annotation/annotation_stellarscope.tsv")␣
↪# generated with annotation_scripts/create_annotations_mouse.Rmd
head(annotation_stellarscope)
```

A data.frame: 6 × 10

| | chr <chr> | start <int> | end <int> | sw_score <int> | strand <chr> | locusID <chr> | family <chr> | subfam <chr> |
|---|---|---|---|---|---|---|---|---|
| 1 | chr1 | 3000001 | 3002128 | 12955 | - | L1_Mus3_dup24 | L1 | L1-Mu |
| 2 | chr1 | 3003153 | 3003994 | 1216 | - | L1Md_F_dup2 | L1 | L1Md- |
| 3 | chr1 | 3003994 | 3004054 | 234 | - | L1_Mus3_dup25 | L1 | L1-Mu |
| 4 | chr1 | 3004041 | 3004206 | 3685 | + | L1_Rod_dup1 | L1 | L1-Roc |
| 5 | chr1 | 3004271 | 3005001 | 3685 | + | L1_Rod_dup2 | L1 | L1-Roc |
| 6 | chr1 | 3005002 | 3005439 | 1280 | + | L1_Rod_dup3 | L1 | L1-Roc |

[ ]:
```
# Remove some classes of TEs (already not present in SoloTE)
classesToExclude <- c("Other", "Satellite", "Unknown", "RNA")
starsoloTEs <- Features(objTE_STARsolo)

filteredStarsoloTEs <-
 ↪annotation_stellarscope[annotation_stellarscope$stellarscopeID %in%
 ↪starsoloTEs, ] %>%
    dplyr::filter(!class %in% classesToExclude) %>%
    pull(stellarscopeID)

print("Percentage of TEs in removed orders:")
print(length(setdiff(starsoloTEs, filteredStarsoloTEs)) / length(starsoloTEs) *
 ↪100)

objTE_STARsolo <- objTE_STARsolo[intersect(starsoloTEs, filteredStarsoloTEs), ]

print("Summary of n counts")
print(summary(objTE_STARsolo$nCount_RNA))
print("Summary of n feature")
print(summary(objTE_STARsolo$nFeature_RNA))

print("Loci present in the annotation:")
print(table(rownames(objTE_STARsolo) %in%
 ↪annotation_stellarscope$stellarscopeID))
```

```
[1] "Percentage of TEs in removed orders:"
[1] 1.308805
[1] "Summary of n counts"
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  244.4  1063.2  1835.9  2439.8  3003.5 43630.4
[1] "Summary of n feature"
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
    402    1670    2140    2250    2687    6974
[1] "Loci present in the annotation:"

 TRUE
37929
```

[ ]:
```
feature_metadata <- annotation_stellarscope[match(Features(objTE_STARsolo),
↪annotation_stellarscope$stellarscopeID),]

head(feature_metadata)
```

A data.frame: 6 × 10

| | chr | start | end | sw_score | strand | locusID | family |
|---|---|---|---|---|---|---|---|
| | <chr> | <int> | <int> | <int> | <chr> | <chr> | <chr> |
| 32 | chr1 | 3031359 | 3031710 | 2885 | - | IAPLTR1a_Mm_dup1 | ERVK |
| 62 | chr1 | 3054839 | 3056749 | 11410 | + | RLTR45-int_dup4 | ERVK |
| 64 | chr1 | 3057320 | 3059492 | 18188 | + | RLTR45-int_dup6 | ERVK |
| 89 | chr1 | 3076927 | 3077419 | 4306 | - | MT2_Mm_dup2 | ERVL |
| 176 | chr1 | 3143093 | 3143442 | 2310 | - | IAPEY2_LTR | ERVK |
| 196 | chr1 | 3155860 | 3156180 | 2776 | + | RLTRETN_Mm_dup2 | ERVK |

[ ]:
```
options(repr.plot.width=7, repr.plot.height=5)


objTE_STARsolo@meta.data$nCount_TE <- objTE_STARsolo@meta.data$nCount_RNA
objTE_STARsolo@meta.data$nFeature_TE <- objTE_STARsolo@meta.data$nFeature_RNA
# Visualize QC metrics as a violin plot
VlnPlot(objTE_STARsolo, features = c("nCount_TE"), ncol = 1,
        cols = colorDict, pt.size = 0) + theme(text=element_text(size=17))

ggsave("figures_10xMouse_RA/nCountTE_violin_STARsolo.png", device='png',dpi=600)
ggsave("figures_10xMouse_RA/nCountTE_violin_STARsolo.pdf", device='pdf')

VlnPlot(objTE_STARsolo, features = c("nFeature_TE"), ncol = 1,
        cols = colorDict, pt.size = 0) + theme(text=element_text(size=17))
ggsave("figures_10xMouse_RA/nFeatureTE_violin_STARsolo.png",
↪device='png',dpi=600)
ggsave("figures_10xMouse_RA/nFeatureTE_violin_STARsolo.pdf", device='pdf')
```

```
Warning message:
"Default search for "data" layer in "RNA" assay yielded no results; utilizing
"counts" layer instead."

Warning message:
"The `slot` argument of `FetchData()` is deprecated as of SeuratObject
5.0.0.
 Please use the `layer` argument instead.
 The deprecated feature was likely used in the Seurat
package.
  Please report the issue at
<https://github.com/satijalab/seurat/issues>."
Warning message:
"`PackageCheck()` was deprecated in SeuratObject 5.0.0.
 Please use `rlang::check_installed()` instead.
 The deprecated feature was likely used in the Seurat
```

```
package.
  Please report the issue at
<https://github.com/satijalab/seurat/issues>."
Warning message:
"No shared levels found between `names(values)` of the manual scale and
the
data's fill values."
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Saving 7 x 7 in image
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```

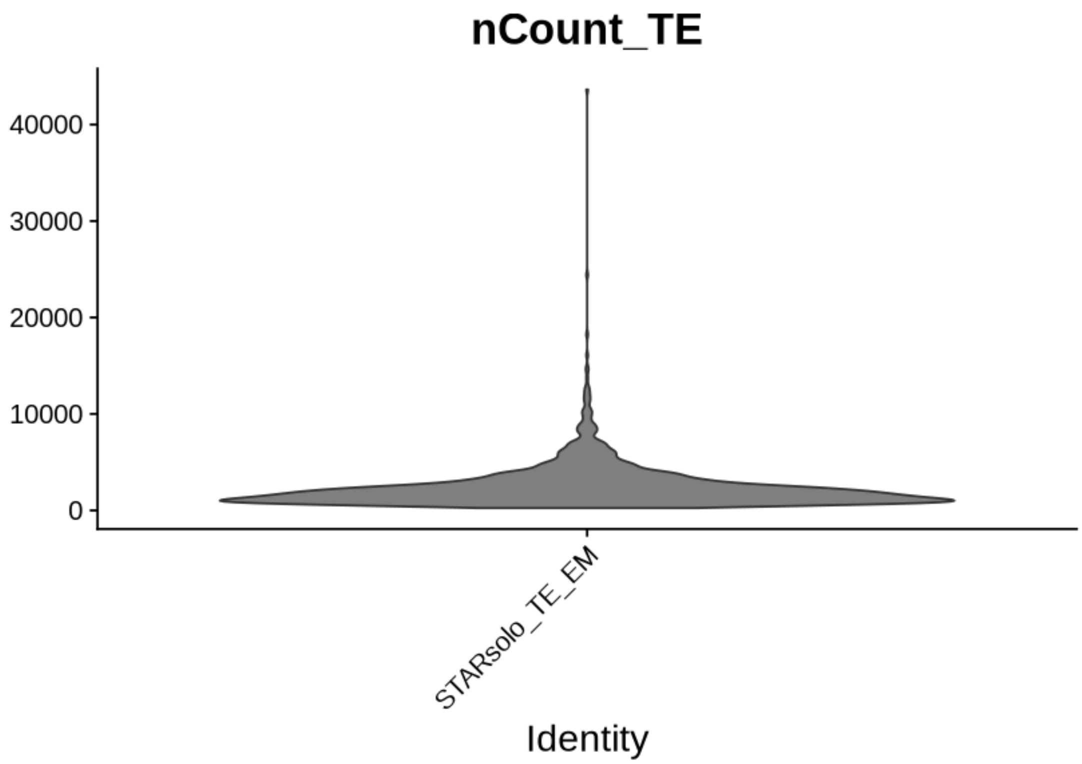


```
Saving 7 x 7 in image
Warning message:
"No shared levels found between `names(values)` of the manual scale and
the
data's fill values."
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"No shared levels found between `names(values)` of the manual scale and
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Saving 7 x 7 in image
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"No shared levels found between `names(values)` of the manual scale and
the
data's fill values."
Warning message:
"No shared levels found between `names(values)` of the manual scale and
the
data's fill values."
```
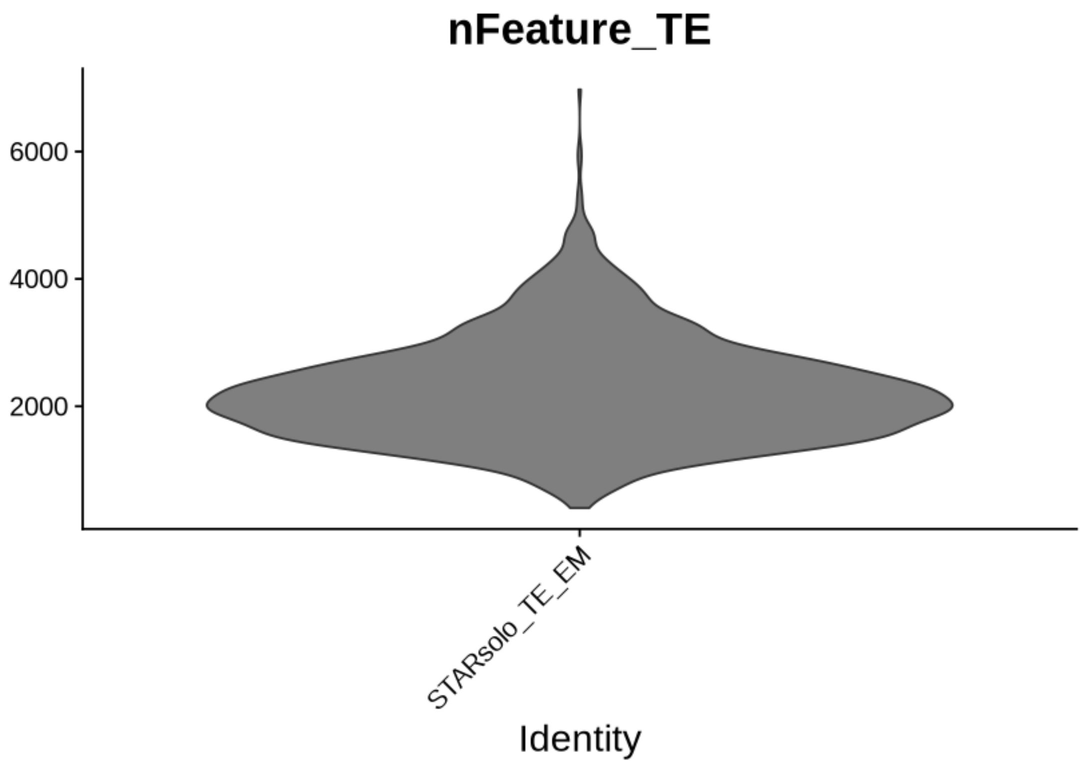




```
[ ]: objTE_STARsolo <- JoinLayers(objTE_STARsolo)

     objTE_STARsolo <- NormalizeData(objTE_STARsolo, normalization.method =␣
      ↪"LogNormalize", scale.factor = 10000)

     objTE_STARsolo <- FindVariableFeatures(objTE_STARsolo, selection.method =␣
      ↪"vst", nfeatures = 4000)

     # Identify the 10 most highly variable genes
     top10 <- head(VariableFeatures(objTE_STARsolo), 10)

     # plot variable features with and without labels
     plot1 <- VariableFeaturePlot(objTE_STARsolo)
     plot2 <- LabelPoints(plot = plot1, points = top10, repel = TRUE)
     plot2
```

```
Normalizing layer: counts

Finding variable features for layer counts
```

[ ]:
```
gc()
all.genes <- rownames(objTE_STARsolo)
objTE_STARsolo <- ScaleData(objTE_STARsolo) # on hvgs
```

A matrix: $2 \times 6$ of type dbl

| | used | (Mb) | gc trigger | (Mb) | max used | (Mb) |
|---|---|---|---|---|---|---|
| Ncells | 14976892 | 799.9 | 42168655 | 2252.1 | 36909475 | 1971.2 |
| Vcells | 133072577 | 1015.3 | 394748854 | 3011.7 | 556827001 | 4248.3 |

```
Centering and scaling data matrix
```

[ ]:
```
grep("MERVL", all.genes, value = T)[1:20]
```

1. 'MERVL-int-dup2' 2. 'MERVL-2A-int-dup12' 3. 'MERVL-int-dup3' 4. 'MERVL-int-dup5' 5. 'MERVL-int-dup8' 6. 'MERVL-int-dup10' 7. 'MERVL-int-dup13' 8. 'MERVL-int-dup19' 9. 'MERVL-int-dup23' 10. 'MERVL-int-dup29' 11. 'MERVL-int-dup30' 12. 'MERVL-int-dup33' 13. 'MERVL-int-dup35' 14. 'MERVL-int-dup50' 15. 'MERVL-int-dup51' 16. 'MERVL-int-dup53' 17. 'MERVL-int-dup56' 18. 'MERVL-int-dup58' 19. 'MERVL-int-dup60' 20. 'MERVL-int-dup63'

[ ]:
```
objTE_STARsolo <- RunPCA(objTE_STARsolo, features = VariableFeatures(object =
  ↪objTE_STARsolo))

DimPlot(objTE_STARsolo, reduction = "pca") + NoLegend()

ElbowPlot(objTE_STARsolo)
```

```
PC_ 1
Positive:  B2-Mm1t-dup17879, B2-Mm1t-dup5682, B2-Mm1a-dup10686, B2-Mm1t-dup5694,
B2-Mm1t-dup5704, B2-Mm2-dup57274, B2-Mm1a-dup1172, B2-Mm1a-dup16080,
B2-Mm1a-dup8415, B2-Mm1a-dup1853
           B2-Mm1t-dup21713, B2-Mm2-dup26127, B2-Mm1a-dup3993, B2-Mm1t-dup21294,
B2-Mm1a-dup11294, B2-Mm1a-dup4262, B2-Mm1a-dup3667, B2-Mm1t-dup6060,
B2-Mm2-dup45338, B2-Mm1a-dup16575
           B2-Mm1t-dup16798, B2-Mm1a-dup14678, B2-Mm1t-dup12961,
B2-Mm1a-dup5340, B2-Mm1a-dup16553, B2-Mm2-dup67657, B2-Mm2-dup62580,
B2-Mm1a-dup9200, B2-Mm1t-dup18041, B4-dup58342
Negative:  L1Md-T-dup18258, RLTR45-int-dup507, RLTR45-dup570, RLTR45-dup757,
RLTR45-dup569, RLTR45-int-dup769, RLTR45-int-dup510, ERVB3-1-I-MM-int-dup14,
ETnERV3-int-dup124, L1Md-T-dup740
           L1Md-T-dup8625, L1Md-T-dup11416, RLTR45-int-dup1196, RLTR4-MM-int-
dup154, LTRIS2-dup341, L1Md-T-dup8184, MMERGLN-int-dup168, L1Md-T-dup4183,
MT2-Mm-dup145, IAPEz-int-dup1770
           L1Md-T-dup10180, L1Md-T-dup4689, B2-Mm1a-dup15511, L1Md-T-dup5818,
L1Md-T-dup22935, L1Md-A-dup16053, RLTR10-dup2494, MT2-Mm-dup46, L1Md-T-dup7347,
L1Md-T-dup20015
PC_ 2
Positive:  MMERVK10C-int-dup1618, MMERVK10C-int-dup929, Lx3-Mus-dup3993,
Lx3-Mus-dup3994, Lx3-Mus-dup3995, B2-Mm1a-dup11536, RLTR6-int-dup1217,
```

MMERVK10C-int-dup636, B2-Mm1a-dup2671, ERVB4-2-LTR-MM-dup54
ORR1A0-int-dup343, B2-Mm1t-dup19755, B2-Mm1a-dup15900, B2-Mm1a-dup12685, B2-Mm1a-dup624, B2-Mm1a-dup10714, B2-Mm1a-dup15899, B2-Mm1t-dup14181, MMERGLN-int-dup734, B2-Mm1a-dup7928
B2-Mm2-dup76118, B2-Mm1t-dup15951, B2-Mm2-dup77197, MMERVK10C-int-dup845, B2-Mm1t-dup1559, B2-Mm1a-dup9287, B2-Mm2-dup56823, B2-Mm1a-dup17634, B2-Mm1a-dup12352, B2-Mm1a-dup3528
Negative: B2-Mm1a-dup14075, B2-Mm1a-dup12940, B2-Mm2-dup79569, B2-Mm1a-dup1664, B2-Mm1a-dup12359, B2-Mm1t-dup17989, B2-Mm1a-dup14692, B2-Mm1a-dup13072, B2-Mm1a-dup11559, B2-Mm1a-dup1082
B2-Mm1a-dup4955, B2-Mm1t-dup16804, B2-Mm2-dup89404, B2-Mm1a-dup10095, B2-Mm1t-dup13658, B2-Mm2-dup5113, B2-Mm1a-dup12114, B2-Mm1t-dup15650, B2-Mm1t-dup22842, B2-Mm2-dup57336
B2-Mm1t-dup15219, B3A-dup70742, B2-Mm1a-dup6097, B2-Mm2-dup84489, B2-Mm2-dup61969, B2-Mm1t-dup17031, B2-Mm1t-dup21169, B2-Mm1t-dup688, B2-Mm2-dup75913, B2-Mm1a-dup10813
PC_ 3
Positive: RLTR31B2-dup143, B2-Mm1a-dup14713, B1-Mus2-dup4086, B2-Mm1t-dup17622, B2-Mm1t-dup17573, B2-Mm1t-dup15936, B2-Mm2-dup55880, B2-Mm2-dup76769, B2-Mm2-dup75064, B2-Mm1a-dup1363
B2-Mm1a-dup9111, B2-Mm2-dup18104, B2-Mm2-dup17723, B2-Mm2-dup19925, B2-Mm1t-dup16919, B2-Mm1a-dup16348, B2-Mm2-dup29363, B2-Mm2-dup17585, B2-Mm1t-dup7087, B2-Mm2-dup23671
B2-Mm1t-dup1763, B1-Mur3-dup17474, B2-Mm1a-dup17210, B2-Mm2-dup26457, B2-Mm2-dup84517, B2-Mm2-dup57584, B2-Mm1a-dup13935, B2-Mm1t-dup5357, B2-Mm1a-dup11824, B2-Mm1t-dup16827
Negative: B2-Mm1a-dup7437, B2-Mm1t-dup22684, B2-Mm1a-dup1180, B2-Mm1a-dup14678, B2-Mm1t-dup22873, B2-Mm1a-dup8925, B2-Mm1a-dup13057, B2-Mm1a-dup850, B2-Mm2-dup38216, B2-Mm1a-dup14225
B2-Mm1a-dup12932, B2-Mm1t-dup20530, B2-Mm1a-dup4097, B2-Mm1t-dup22763, B2-Mm1a-dup14547, B2-Mm1t-dup6183, B2-Mm1t-dup9010, B2-Mm1a-dup13029, B2-Mm1t-dup13064, B2-Mm1a-dup3943
B2-Mm1a-dup14872, B2-Mm1a-dup14973, B2-Mm1a-dup16206, B2-Mm1a-dup4969, B2-Mm1a-dup5816, B2-Mm2-dup19836, B2-Mm1a-dup6037, B2-Mm1a-dup2117, B2-Mm1a-dup5573, B2-Mm1a-dup13426
PC_ 4
Positive: B2-Mm1a-dup1592, B2-Mm1t-dup9736, B2-Mm1t-dup17140, B2-Mm1t-dup16914, B2-Mm1a-dup13057, B2-Mm1t-dup15202, B2-Mm1a-dup12932, B2-Mm2-dup29106, B2-Mm1t-dup3859, B2-Mm1a-dup6854
B1-Mus2-dup19471, B2-Mm2-dup76118, B2-Mm2-dup20705, B2-Mm1a-dup1180, B2-Mm2-dup49518, B2-Mm1a-dup5943, B2-Mm1t-dup18299, B2-Mm1a-dup1571, B2-Mm2-dup82343, B2-Mm1t-dup11904
B2-Mm1t-dup9878, B2-Mm1a-dup10695, B2-Mm1a-dup17990, B2-Mm1a-dup6903, B2-Mm2-dup30203, B2-Mm1t-dup12949, B2-Mm1a-dup2872, B2-Mm1t-dup19750, B2-Mm1a-dup3666, B2-Mm1a-dup17786
Negative: B2-Mm1a-dup8968, B2-Mm1t-dup21020, B2-Mm1a-dup11646, B2-Mm1t-dup10003, B2-Mm1a-dup1594, B2-Mm1a-dup1873, B2-Mm1a-dup5689, B2-Mm2-dup24388, B2-Mm1t-dup17080, B2-Mm1a-dup1189

B2-Mm1t-dup14807, B2-Mm1a-dup9265, B2-Mm1a-dup660, B2-Mm1a-dup5714, B2-Mm2-dup40705, B2-Mm1a-dup5915, B2-Mm1a-dup14012, B2-Mm2-dup73048, B1-Mus2-dup32414, B2-Mm1t-dup868

B2-Mm1a-dup8396, B2-Mm1t-dup12513, B2-Mm1a-dup428, B2-Mm1a-dup10814, B2-Mm2-dup17739, B2-Mm1a-dup1443, B2-Mm1t-dup6060, B2-Mm1a-dup16240, B2-Mm1a-dup3667, B2-Mm1t-dup15428

PC_ 5

Positive: B2-Mm1a-dup4262, B2-Mm1a-dup25, B2-Mm1t-dup9570, B2-Mm1t-dup1099, B2-Mm1a-dup12322, B2-Mm1a-dup9203, B2-Mm1a-dup14225, B2-Mm1a-dup3933, B2-Mm2-dup33703, B2-Mm1a-dup4188

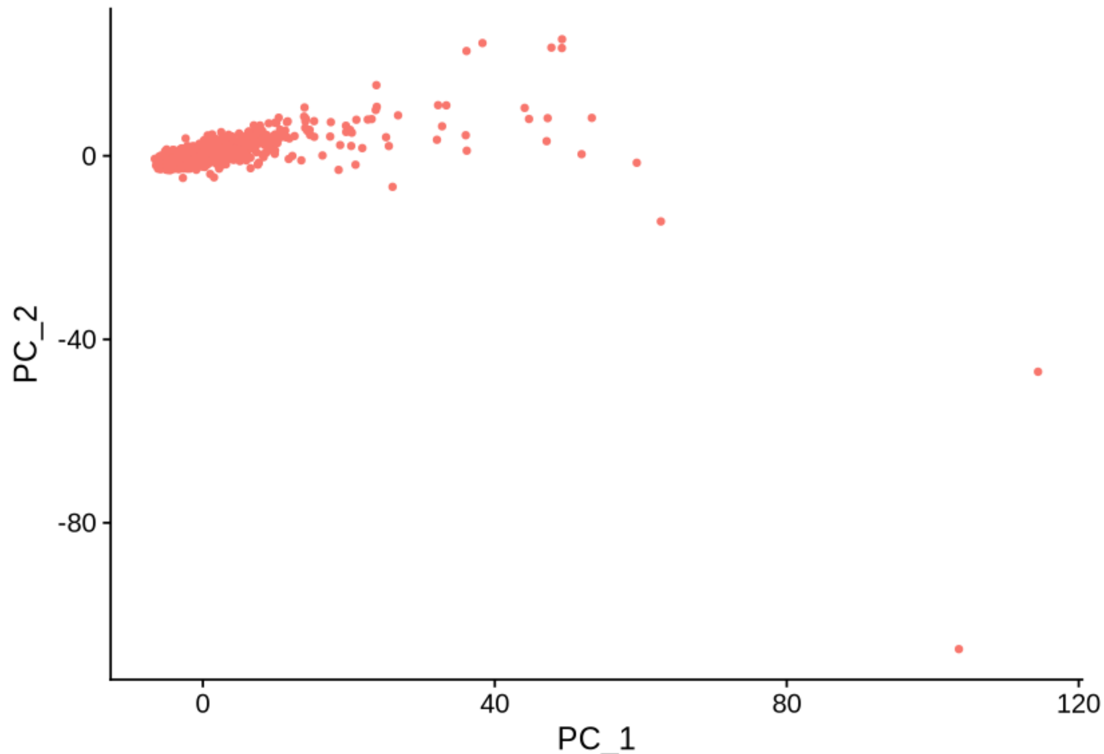
B2-Mm2-dup68353, B2-Mm1t-dup256, B2-Mm1t-dup16966, B2-Mm1t-dup13631, B2-Mm2-dup67657, B2-Mm1a-dup2155, B2-Mm1a-dup6716, B2-Mm1t-dup5047, B2-Mm1a-dup6352, B2-Mm2-dup713

B2-Mm1a-dup14324, B2-Mm2-dup21060, B2-Mm2-dup5113, B2-Mm1a-dup12014, B2-Mm1a-dup8415, B2-Mm1a-dup15900, B2-Mm1t-dup16914, B2-Mm2-dup53945, B2-Mm1t-dup7323, B2-Mm1a-dup12360

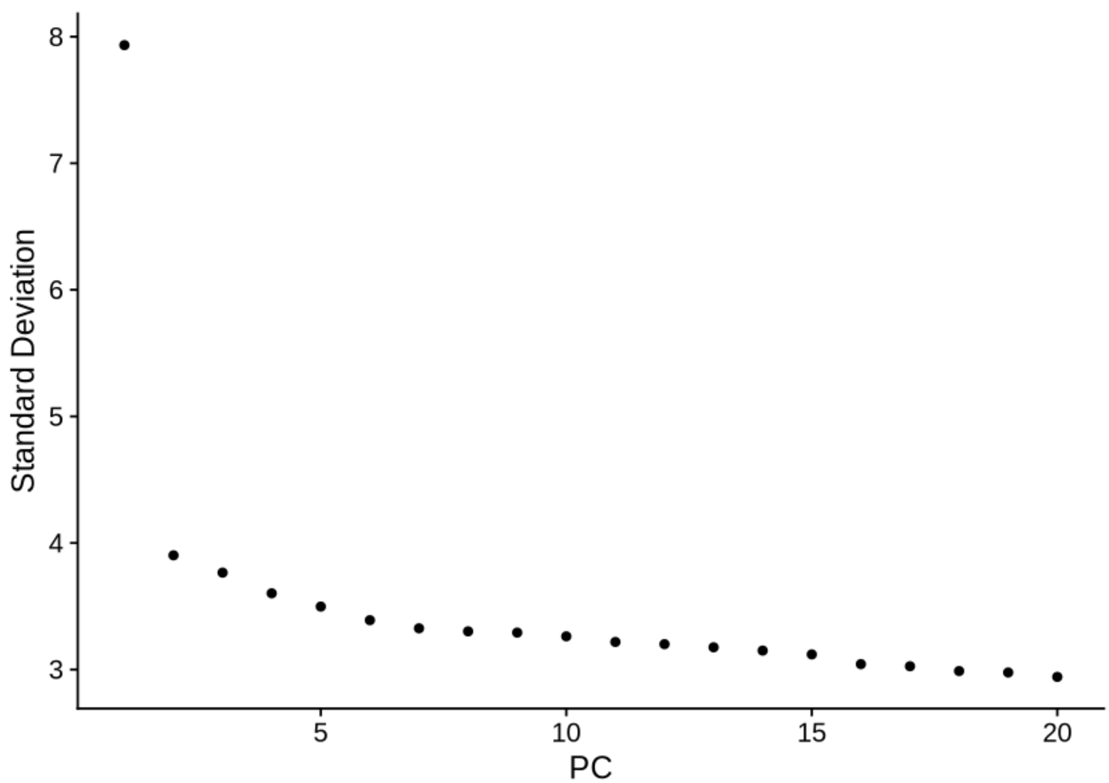
Negative: B2-Mm1a-dup1167, B2-Mm1a-dup6980, B2-Mm2-dup21298, B2-Mm1a-dup14004, B2-Mm1t-dup18305, B2-Mm1t-dup14492, B2-Mm2-dup61557, B2-Mm1a-dup12827, B2-Mm1t-dup3602, B2-Mm2-dup78680

B2-Mm1t-dup1657, B2-Mm2-dup43758, IAPEz-int-dup4789, B2-Mm1t-dup21346, B1-Mus1-dup8091, B2-Mm1t-dup20486, B2-Mm2-dup12576, B2-Mm1a-dup2429, B2-Mm1a-dup1314, B2-Mm1a-dup10141

B2-Mm1t-dup21574, B2-Mm1t-dup6402, B2-Mm1t-dup21737, B2-Mm2-dup19446, B2-Mm1a-dup15836, B2-Mm1t-dup16410, B2-Mm1a-dup13246, B2-Mm1a-dup4190, B2-Mm1a-dup14872, B2-Mm1t-dup4951



0
-40
-80
PC_2
0
40
80
120
PC_1

8
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Standard Deviation
5
10
15
20
PC

[ ]:
```
objTE_STARsolo <- FindNeighbors(objTE_STARsolo, dims = 1:12, k.param = 20)
objTE_STARsolo <- FindClusters(objTE_STARsolo, resolution = 1)
objTE_STARsolo <- RunUMAP(objTE_STARsolo, dims = 1:12)
DimPlot(objTE_STARsolo, reduction = "umap")
```

```
Computing nearest neighbor graph


Computing SNN


Modularity Optimizer version 1.3.0 by Ludo Waltman and Nees Jan van Eck

Number of nodes: 1507
Number of edges: 57305

Running Louvain algorithm…
Maximum modularity in 10 random starts: 0.7129
Number of communities: 4
Elapsed time: 0 seconds
14:54:02 UMAP embedding parameters a = 0.9922 b = 1.112

14:54:02 Read 1507 rows and found 12 numeric columns

14:54:02 Using Annoy for neighbor search, n_neighbors = 30

14:54:02 Building Annoy index with metric = cosine, n_trees = 50

0%   10   20   30   40   50   60   70   80   90   100%

[----|----|----|----|----|----|----|----|----|----|

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14:54:02 Writing NN index file to temp file
/mnt/TEdataStorage_2T/tmp/Rtmp2sOVvR/file351cd833cc121

14:54:03 Searching Annoy index using 1 thread, search_k = 3000

14:54:03 Annoy recall = 100%

14:54:05 Commencing smooth kNN distance calibration using 1 thread
 with target n_neighbors = 30
```
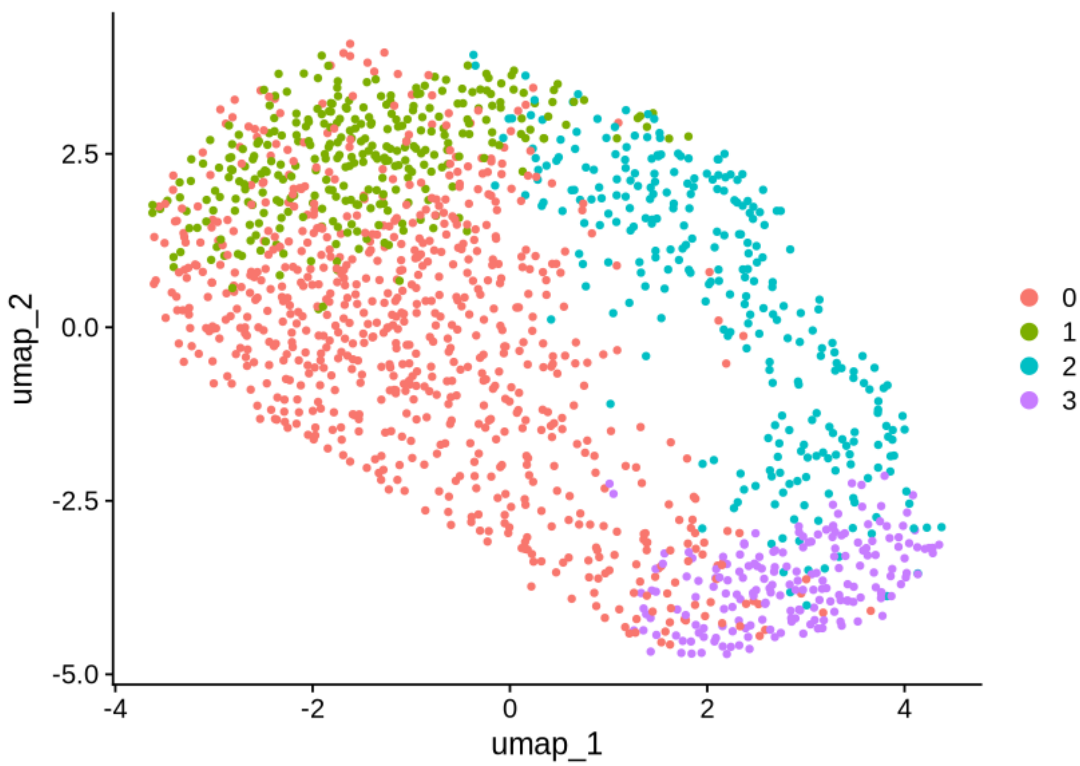
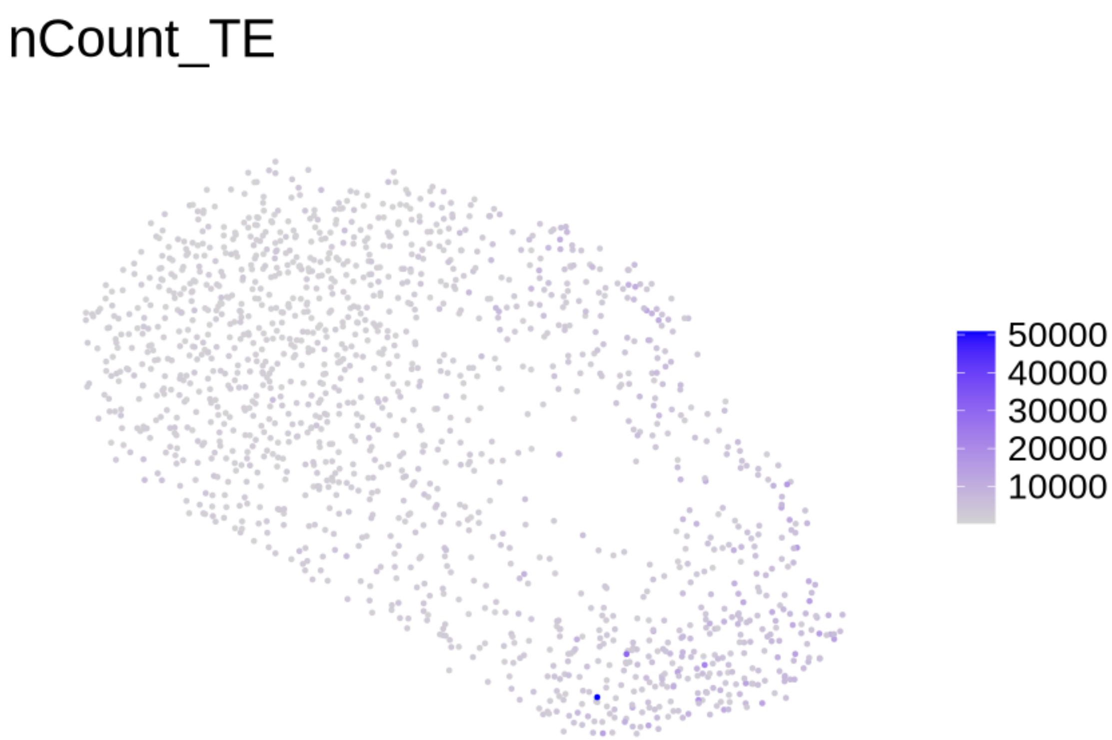
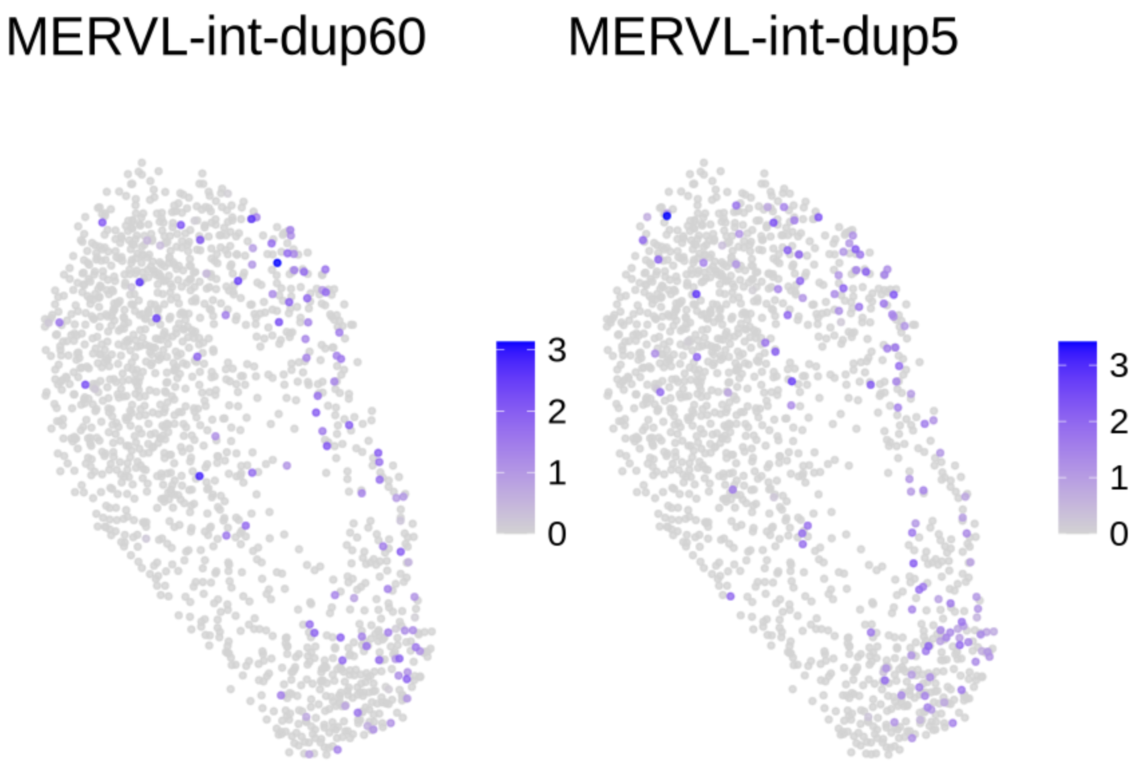

```
14:54:08 Initializing from normalized Laplacian + noise (using RSpectra)

14:54:08 Commencing optimization for 500 epochs, with 67648 positive edges

14:54:08 Using rng type: pcg

14:54:12 Optimization finished
```




[ ]:
```
FeaturePlot(objTE_STARsolo, reduction = "umap", features = "nCount_TE", pt.size
 ↪= 0.5) +
  theme_void() +
  theme(text=element_text(size=20))
```




```
[ ]: FeaturePlot(objTE_STARsolo, features = c('MERVL-int-dup60','MERVL-int-dup5'),
     ↪pt.size=1,alpha = 0.8,order = TRUE ) &
      theme_void() &
      theme(text=element_text(size=20))
```




[ ]:
```
saveRDS(objTE_STARsolo, "data_10xMouse_RA/STARsolo_10xMouse_RA_seuratObj_251125.
↪RDS")
```

[ ]:
```

```